PRACTICAL-4

1.Aim:

Interface GPS module with Raspberry Pi

2.Prior Concepts

i)Concept of Python Programming

ii) Working of GPS Module Sensor

3.New Concepts

i) Hardware connection of GPS Module with Raspi.

4.Hardware Requirement and PIN Connection

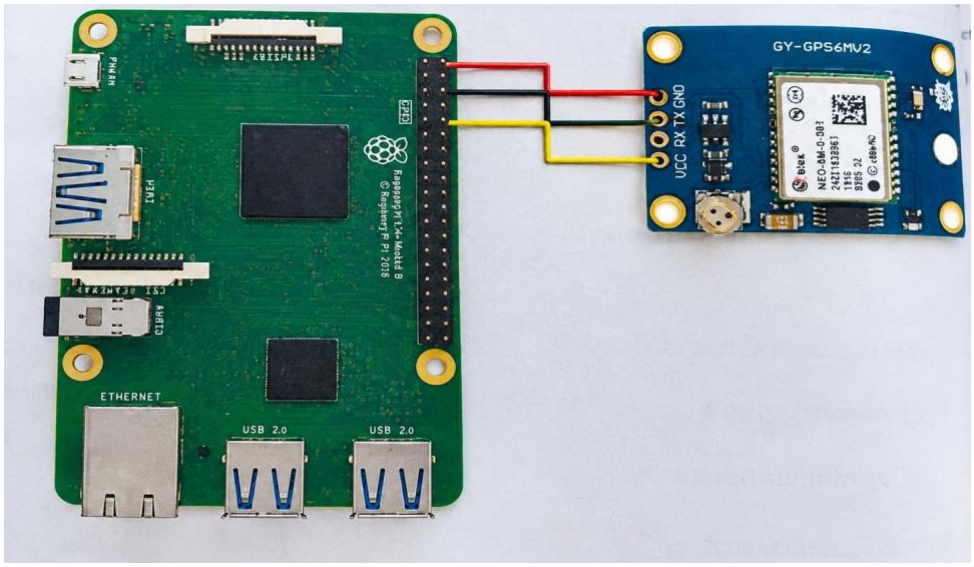
Raspberry Pi 4

ii) GPS Module (e.g., NEO-6M GPS Module)

iii) Jumper Wires

Pin Connection

GPS Module PIN	RPi	Function
GND(Ground)	6	GND(Ground)
VCC(+5V Power)	2	5V
Tx	10	GPIO 15(RXD)
Rx	8	GPIO 14(TXD)



5. Procedure

i) Enable UART on Raspberry Pi by writing below code in command terminal

sudo raspi-config

ii)Navigate:

Interface Options

→ Serial Port

→ Login shell over serial? No

→ Serial port hardware enabled? Yes

iii) Reboot Raspi by writing below code in command terminal

sudo reboot

iv) Verify Serial port by writing below code in command terminal

Is /dev/serial*

Expected Output:

/dev/serial0

/dev/serial1

v) Install GPS Packages, on Raspberry Pi by writing below code in command terminal

sudo apt update

sudo apt install gpsd gpsd-clients python3-gps -y

vi) Stop GPS daemon, on Raspberry Pi by writing below code in command terminal sudo systemctl stop gpsd.socket

sudo systemctl disable gpsd.socket

vii)Start GPS daemon manually, on Raspberry Pi by writing below code in command terminal.

sudo gpsd /dev/serial0-F/var/run/gpsd.sock

viii) Test GPS Data, on Raspberry Pi by writing below code in command terminal

cgps-s

OR

gpsmon

Note: You should see NMEA sentences and GPS coordinates once the module acquires satellites.

Example:

Latitude: 19.0760

Longitude: 72.8777

Altitude: 14.2 m

Time: 12:45:30 UTC

ix) Write the below command to read data from GPS.

pip3 install pyserial pynmea2

Write the below code:

```
import serial
import pynmea2

port = "/dev/serial0"
ser = serial.Serial(port, baudrate=9600, timeout=1)

while True:
    data = ser.readline().decode('ascii', errors='replace')

if data.startswith('$GPGGA'):
    msg = pynmea2.parse(data)
    print("Latitude:", msg.latitude)
```

```
print("Longitude:", msg.longitude)
 print("Altitude :", msg.altitude)
 print("--------------------------")
```

6)Results: